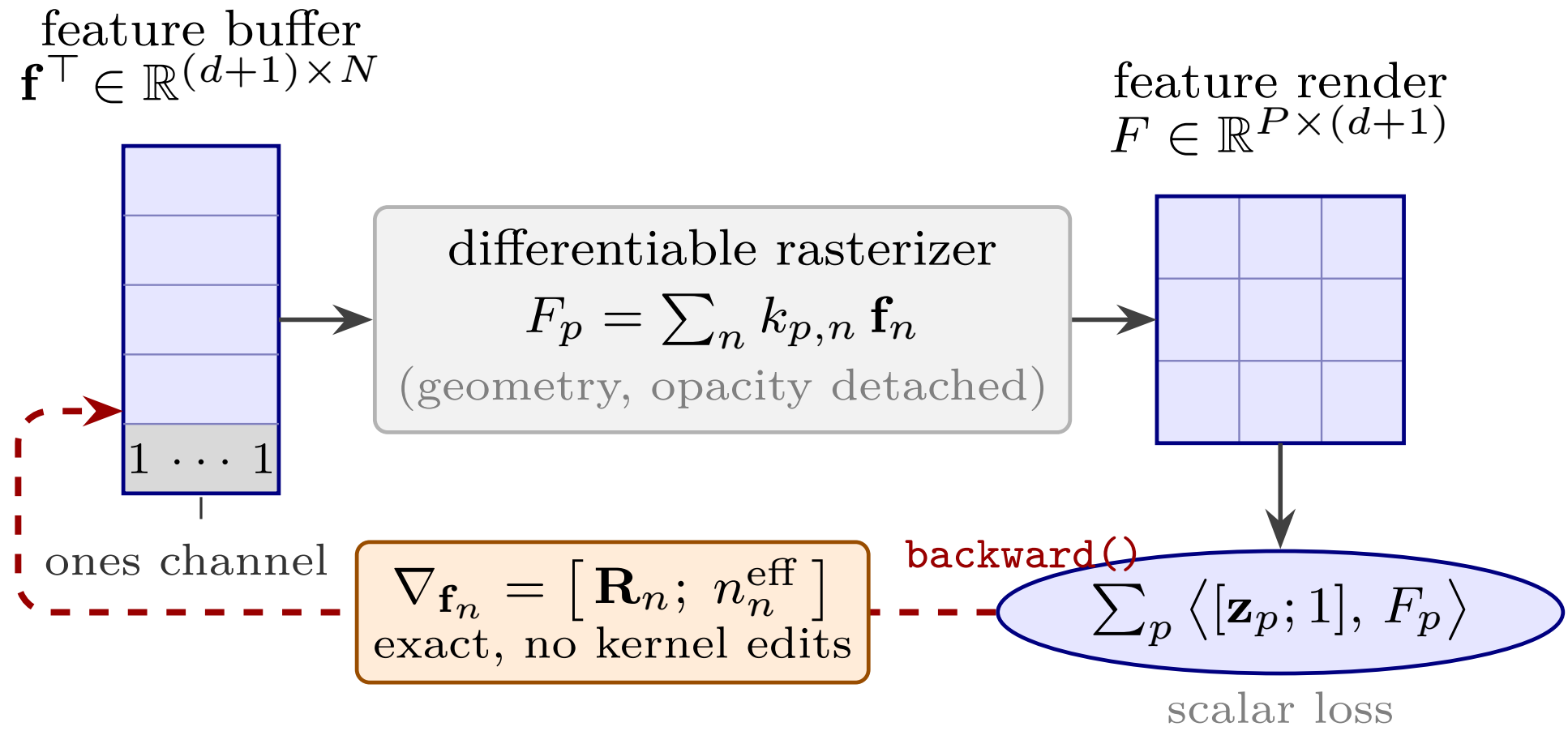
feature buffer
$\mathbf{f}^\top \in \mathbb{R}^{(d+1)\times N}$
feature render
$F \in \mathbb{R}^{P\times(d+1)}$
differentiable rasterizer
$F_p = \sum_n k_{p,n}\,\mathbf{f}_n$
(geometry, opacity detached)
$1 \cdots 1$
ones channel
backward()
$\nabla_{\mathbf{f}_n} = [\,\mathbf{R}_n;\; n_n^{\text{eff}}\,]$
exact, no kernel edits
$\sum_p \langle [\mathbf{z}_p; 1],\, F_p \rangle$
scalar loss
one render + one backward per keyframe, any feature rasterizer